Lab: Adding Users and Groups to Azure Databricks and Assigning Entitlements

Description

You are the administrator for a new data team. You will create a Data Engineers group and a Data Analysts group in Microsoft Entra ID, bring both groups into the Databricks account, give them access to the workspace, and assign each group different entitlements.

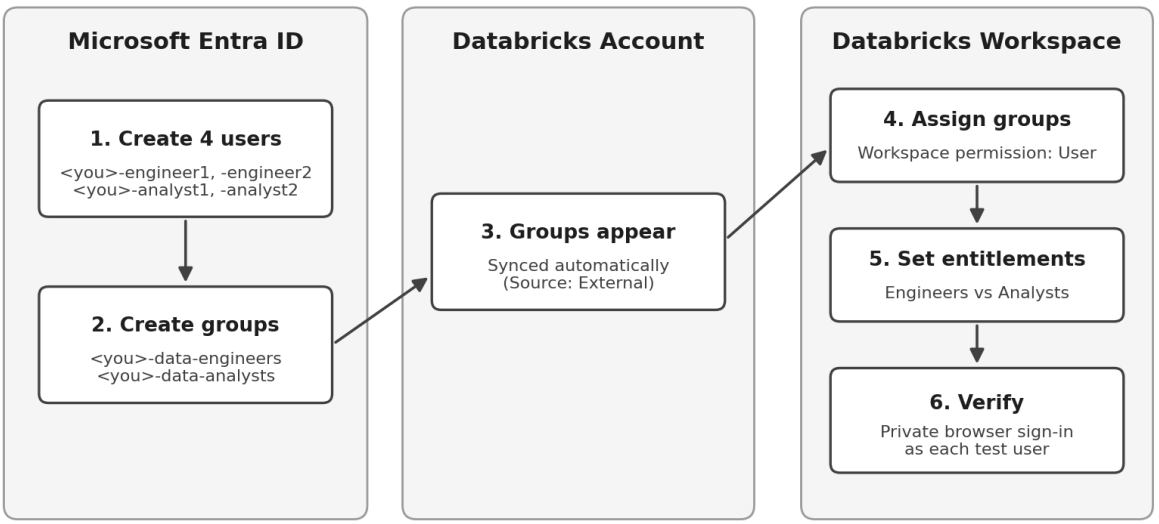
You will then sign in as a test user from each group to see the difference for yourself.

Learning Objectives

- Create users and security groups in Microsoft Entra ID
- Add Entra ID groups to the Databricks account using automatic identity management
- Assign groups to a Databricks workspace
- Grant different entitlements to Data Engineers and Data Analysts
- Verify access by signing in as a non-admin user

Step by Step Guide

Lab flow



Your lab values

Your **prefix** is your first name in lowercase. Use it in every name you create, so your groups do not clash with other learners' groups.

Your account	Prefix
Andrew Learner	andrew
Bob Learner	bob
Carol Learner	carol
Dave Learner	dave
Saurabh Learner	saurabh
Yu Learner	yu

Item	Value
Tenant domain	saurabhuptut.onmicrosoft.com
Databricks account console	https://accounts.azuredatabricks.net
Workspace URL	Provided by your trainer

Step 1: Check your access

1. Sign in to the Azure portal (portal.azure.com) with your learner account.
2. In another tab, open the Databricks account console and sign in with the same account.
3. In the account console, go to **Security** > **User provisioning**.
4. Look at Automatic identity management. Do not change it.

> **Check:** Automatic identity management shows Enabled. This is what lets Databricks see your Entra ID groups.

Step 2: Create four internal users in Entra ID

These are new internal users in your tenant with no admin roles, so they show you exactly what a normal team member sees.

User principal name	Display name	Will join
<prefix>-engineer1	<Prefix> Engineer 1	data-engineers
<prefix>-engineer2	<Prefix> Engineer 2	data-engineers

User principal name	Display name	Will join
<prefix>-analyst1	<Prefix> Analyst 1	data-analysts
<prefix>-analyst2	<Prefix> Analyst 2	data-analysts

1. In the Azure portal, open Microsoft Entra ID > Users > New user > Create new user. Do not use Invite external user.
2. Enter the user principal name and display name from the table.
3. Keep Auto-generate password selected. Copy the passwords for engineer1 and analyst1. You will sign in as them later.
4. Do not assign any roles. Click Review + create > Create.
5. Repeat until all four users are created.

> Check: All four users appear in the Users list with User type Member.

Step 3: Create two security groups in Entra ID

1. Open Microsoft Entra ID > Groups > New group.
2. Group type: Security. Group name: <prefix>-data-engineers. Membership type: Assigned.
3. Under Members, add <prefix>-engineer1 and <prefix>-engineer2. Click Create.
4. Repeat for <prefix>-data-analysts. Add <prefix>-analyst1 and <prefix>-analyst2.

> Check: Open each group and confirm it has 2 members.

Note: Group membership is always managed in Entra ID. Databricks reads it but cannot edit it.

Step 4: Add the groups to the Databricks account

1. In the account console, go to User management > Groups.
2. Click Add group, search for <prefix>-data-engineers, select it and add it.
3. Repeat for <prefix>-data-analysts.
4. Open one of the groups and look at its details.

> Check: Both groups show Source External. The status shows Inactive: No usage, because the group is in the account but not yet in any workspace.

Step 5: Assign the groups to the workspace

1. In the account console, go to Workspaces and open the lab workspace.
2. Open the Permissions tab and click Add permissions.
3. Search for <prefix>-data-engineers and set the permission to User. Save.

4. Repeat for <prefix>-data-analysts, also as User.

> Check: Both groups are listed on the workspace Permissions tab. Back under User management > Groups, their status is now Active.

Step 6: Set entitlements for each group

1. Open the workspace URL and sign in with your learner account.
2. Click your username (top right) > Settings > Identity and access.
3. Next to Groups, click Manage and open <prefix>-data-engineers.
4. Open the Entitlements tab and set the toggles as in the table below. Repeat for <prefix>-data-analysts.

Entitlement	data-engineers	data-analysts
Workspace access	On	Off
Databricks SQL access	On	On
Allow unrestricted cluster creation	On	Off
Consumer access	Off	Off

Note: If the workspace asks you to choose entitlements while adding a group, use the same table.

> Check: Each group shows only the entitlements in the table.

Step 7: Verify as the Data Engineer

1. Open a private browser window (InPrivate in Edge, Incognito in Chrome).
2. Go to the workspace URL and sign in as <prefix>-engineer1@saurabhuptut.onmicrosoft.com with the password from Step 2.
3. Set a new password when asked. Complete MFA setup if prompted.
4. Click New > Notebook to check you can create a notebook.
5. Open Compute and look for the Create compute button. You do not need to start a cluster.
6. Close the private window.

> Check: The engineer can create notebooks and compute, and can also open SQL Editor.

Step 8: Verify as the Data Analyst

1. Open a new private browser window.
2. Sign in as <prefix>-analyst1@saurabhuptut.onmicrosoft.com and set a new password.
3. Open SQL Editor and Dashboards.

4. Try to create a notebook and try to create compute.
5. Close the private window.

> Check: The analyst can use SQL Editor, Queries and Dashboards, but cannot create notebooks or clusters.

Step 9 (Optional): Change an entitlement and re-test

1. As your learner account, turn **Workspace access** On for <prefix>-data-analysts.
2. Sign in again as <prefix>-analyst1 in a new private window. Notebooks are now available.
3. Turn **Workspace access** Off again for the analysts group.

> Check: One change on the group changed access for every member of that group.

Outcomes

- Four internal users and two security groups exist in Entra ID with your prefix
- Both groups are in the Databricks account as External groups and assigned to the workspace
- Data Engineers can build notebooks and compute; Data Analysts are limited to SQL
- You have seen that access is managed by group, not by individual user